$3d^5$

August 18, 2026

1 $\left|3d_{xz}^2 3d_{yz}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha 3d_{xy}^0\right\rangle$

Here are the CSFs:

$$
\begin{aligned}
|1,5/2,5/2\rangle &= \left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
|2,5/2,3/2\rangle &= \frac{1}{\sqrt{5}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{5}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{\sqrt{5}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{5}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{\sqrt{5}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
|3,5/2,1/2\rangle &= \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|4,5/2,-1/2\rangle =& \frac{1}{\sqrt{10}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{10}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|5,5/2,-3/2\rangle =& \frac{1}{\sqrt{5}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{5}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{5}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{5}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{5}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
|6,5/2,-5/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle
$$

$$
\begin{aligned}
|7,3/2,3/2\rangle =& +\frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|8,3/2,1/2\rangle =& + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|9,3/2,-1/2\rangle =& + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|10,3/2,-3/2\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
|11,3/2,3/2\rangle =& + \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$\begin{aligned}
\left|12, 3/2, 1/2\right\rangle = &+ \frac{2}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{2}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{2}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}$$

$$\begin{aligned}
\left|13, 3/2, -1/2\right\rangle = &\frac{2}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{2}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{2}{3\sqrt{2}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}$$

$$\begin{aligned}
\left|14, 3/2, -3/2\right\rangle = &\frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle
\end{aligned}$$

$$\begin{aligned}
\left|15, 3/2, 3/2\right\rangle = &+ \frac{\sqrt{3}}{2} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}$$

$$\begin{aligned}
\left|16, 3/2, 1/2\right\rangle = &\frac{1}{2} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{6} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{6} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{6} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{3} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{3} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}$$

$$
\begin{aligned}
|17, 3/2, -1/2\rangle = & \frac{1}{3}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{3}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& - \frac{1}{3}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{6}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle \\
& + \frac{1}{3}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle - \frac{1}{3}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& + \frac{1}{6}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{3}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& + \frac{1}{6}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{2}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|18, 3/2, -3/2\rangle = & \frac{1}{\sqrt{12}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{12}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& + \frac{1}{\sqrt{12}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle - \frac{\sqrt{3}}{2}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|19, 3/2, 3/2\rangle = & \frac{2}{\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle - \frac{1}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle \\
& - \frac{1}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{1}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle \\
& - \frac{1}{2\sqrt{5}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|20, 3/2, 1/2\rangle = & \frac{\sqrt{3}}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{\sqrt{3}}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle + \frac{\sqrt{3}}{2\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle \\
& - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle \\
& + \frac{\sqrt{3}}{2\sqrt{5}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle \\
& - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle - \frac{\sqrt{3}}{3\sqrt{5}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{\alpha}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|21, 3/2, -1/2\rangle = & \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
& + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle + \frac{\sqrt{3}}{3\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{\sqrt{3}}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|22, 3/2, -3/2\rangle = & \frac{1}{2\sqrt{5}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& + \frac{1}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{2\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \frac{2}{\sqrt{5}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle
\end{aligned}
$$

$$|23, 3/2, 3/2\rangle = \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|24, 3/2, 3/2\rangle = \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|25, 3/2, 3/2\rangle = \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|26, 3/2, 3/2\rangle = \left| 3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|27, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|28, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|29, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|30, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|31, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|32, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|33, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2} \right\rangle$$

$$|34, 3/2, 3/2\rangle = \left| 3d_{xz}^{\alpha} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|35,3/2,3/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|36,3/2,3/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|37,3/2,3/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|38,3/2,3/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|39,3/2,3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|40,3/2,3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|41,3/2,3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|42,3/2,3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|43,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|44,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|45,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|46,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|47,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|48,3/2,1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$\begin{aligned}|49,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|50,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\end{aligned}$$

$$\begin{aligned}|51,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|52,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle\end{aligned}$$

$$\begin{aligned}|53,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|54,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|55,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\end{aligned}$$

$$\begin{aligned}|56,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|57,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &+ \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$
\begin{aligned}
|58,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
|59,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|60,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|61,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|62,3/2,1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle\\
|63,3/2,-1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle\\
|64,3/2,-1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|65,3/2,-1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle\\
|66,3/2,-1/2\rangle =&\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\\
&+\frac{1}{\sqrt{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$|67,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\beta}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{\beta}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{\alpha}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|68,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{0}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{\beta}_{xy}3d^{0}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|69,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{2}_{yz}3d^{0}_{xy}3d^{\beta}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{0}_{xy}3d^{\alpha}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{2}_{yz}3d^{0}_{xy}3d^{\beta}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|70,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{2}_{xy}3d^{\beta}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{\alpha}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|71,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{0}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{2}_{xy}3d^{0}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{2}_{xy}3d^{0}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|72,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{2}_{z^2}3d^{0}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{2}_{z^2}3d^{0}_{x^2-y^2}\right\rangle$$

$$|73,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{\beta}_{xy}3d^{0}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{\beta}_{xy}3d^{0}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{\alpha}_{xy}3d^{0}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|74,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{0}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{0}_{xy}3d^{2}_{z^2}3d^{\beta}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{0}_{xy}3d^{2}_{z^2}3d^{\alpha}_{x^2-y^2}\right\rangle$$

$$|75,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d^{\alpha}_{xz}3d^{\beta}_{yz}3d^{0}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\alpha}_{yz}3d^{0}_{xy}3d^{\beta}_{z^2}3d^{2}_{x^2-y^2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d^{\beta}_{xz}3d^{\beta}_{yz}3d^{0}_{xy}3d^{\alpha}_{z^2}3d^{2}_{x^2-y^2}\right\rangle$$

$$|76,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|77,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|78,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|79,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|80,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|81,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|82,3/2,-1/2\rangle = \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{3}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|83,3/2,-3/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|84,3/2,-3/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|85,3/2,-3/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|86,3/2,-3/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|87,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|88,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|89,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|90,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|91,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|92,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|93,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|94,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|95,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|96,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|97,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|98,3/2,-3/2\rangle = \left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$|99,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|100,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|101,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|102,3/2,-3/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle$$

$$\begin{aligned}|103,1/2,1/2\rangle = &+\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\ &- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle\end{aligned}$$

$$\begin{aligned}|104,1/2,-1/2\rangle = &+\frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{2}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\ &- \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{2}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle\end{aligned}$$

$$
\begin{aligned}
|105, 1/2, 1/2\rangle = &+ \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|106, 1/2, -1/2\rangle = &+ \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|107, 1/2, 1/2\rangle = &+ \frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
|108, 1/2, -1/2\rangle = &\frac{1}{\sqrt{3}} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{\sqrt{12}} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{3}} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
|109, 1/2, 1/2\rangle = &+ \frac{2}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{3} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{6} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{6} \left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{6} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{6} \left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{3} \left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|110, 1/2, -1/2\rangle =& \frac{1}{3}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{6}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{1}{6}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{6}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{6}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&+ \frac{2}{3}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|111, 1/2, 1/2\rangle =& \frac{1}{\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|112, 1/2, -1/2\rangle =& \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle \\
&+ \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle \\
&- \frac{1}{3\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle + \frac{1}{\sqrt{2}}\left|3d_{xz}^{\beta}3d_{yz}^{\beta}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
\end{aligned}
$$

$$
|113, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle
$$

$$
|114, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle
$$

$$
|115, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
$$

$$
|116, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{2}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle
$$

$$|117, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|118, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|119, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|120, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|121, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|122, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|123, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2} \right\rangle$$

$$|124, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|125, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle$$

$$|126, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|127, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|128, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle$$

$$|129, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|130, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|131, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle$$

$$|132, 1/2, 1/2\rangle = +\frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle$$

$$|133, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle$$

$$|134, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle$$

$$|135, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|136, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|137, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|138, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|139, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|140, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|141, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|142, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|143, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|144, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|145, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|146, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|147, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|148, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|149, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|150, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|151, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|152, 1/2, -1/2\rangle = \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{2}} \left|3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|153,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|154,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|155,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|156,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{2}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|157,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|158,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|159,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|160,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|161,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|162,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|163,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|164,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|165,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{\beta}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|166,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|167,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|168,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{\beta}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|169,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|170,1/2,1/2\rangle = \sqrt{\frac{2}{3}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\beta}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{\beta}3d_{x^2-y^2}^{\alpha}\right\rangle - \frac{1}{\sqrt{6}}\left|3d_{xz}^{0}3d_{yz}^{\beta}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$
\begin{aligned}
|171, 1/2, 1/2\rangle =& \sqrt{\frac{2}{3}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta} \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
& - \frac{1}{\sqrt{6}} \left| 3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|172, 1/2, 1/2\rangle =& \sqrt{\frac{2}{3}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2} \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^{0} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
& - \frac{1}{\sqrt{6}} \left| 3d_{xz}^{0} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2} \right\rangle \\
|173, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle \\
|174, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|175, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|176, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|177, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0} \right\rangle \\
|178, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha} \right\rangle \\
|179, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\alpha} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta} \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta} \right\rangle \\
& - \sqrt{\frac{2}{3}} \left| 3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha} \right\rangle
\end{aligned}
$$

$$
\begin{aligned}
|180, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{0}\right\rangle \\
|181, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|182, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle \\
|183, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{\alpha} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle \\
|184, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|185, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{\alpha} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle \\
|186, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|187, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\alpha}\right\rangle \\
|188, 1/2, -1/2\rangle =& \frac{1}{\sqrt{6}} \left|3d_{xz}^{\alpha} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle + \frac{1}{\sqrt{6}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\alpha} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle \\
& - \sqrt{\frac{2}{3}} \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{\alpha} 3d_{x^2-y^2}^{2}\right\rangle
\end{aligned}
$$

$$|189,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|190,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|191,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|192,1/2,-1/2\rangle = \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle + \frac{1}{\sqrt{6}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle - \sqrt{\frac{2}{3}}\left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|193,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|194,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0\right\rangle$$

$$|195,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|196,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^0\right\rangle$$

$$|197,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|198,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|199,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0\right\rangle$$

$$|200,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|201,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$

$$|202,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$

$$|203,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$

$$|204,1/2,1/2\rangle = \left|3d_{xz}^2 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$

$$|205,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|206,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|207,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|208,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|209,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|210,1/2,1/2\rangle = \left|3d_{xz}^{\alpha}3d_{yz}^{0}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|211,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{0}\right\rangle$$

$$|212,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|213,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|214,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{\alpha}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|215,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|216,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|217,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{0}\right\rangle$$

$$|218,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{2}3d_{z^2}^{0}3d_{x^2-y^2}^{2}\right\rangle$$

$$|219,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{\alpha}3d_{xy}^{0}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|220,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{2}3d_{x^2-y^2}^{\alpha}\right\rangle$$

$$|221,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{2}3d_{z^2}^{\alpha}3d_{x^2-y^2}^{2}\right\rangle$$

$$|222,1/2,1/2\rangle = \left|3d_{xz}^{0}3d_{yz}^{0}3d_{xy}^{\alpha}3d_{z^2}^{2}3d_{x^2-y^2}^{2}\right\rangle$$

$$|223,1/2,-1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{\beta}3d_{z^2}^{0}3d_{x^2-y^2}^{0}\right\rangle$$

$$|224,1/2,-1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{\beta}3d_{x^2-y^2}^{0}\right\rangle$$

$$|225,1/2,-1/2\rangle = \left|3d_{xz}^{2}3d_{yz}^{2}3d_{xy}^{0}3d_{z^2}^{0}3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|226, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|227, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|228, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{\beta} 3d_{xy}^{0} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|229, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|230, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|231, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|232, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|233, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|234, 1/2, -1/2\rangle = \left|3d_{xz}^{2} 3d_{yz}^{0} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|235, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|236, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|237, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|238, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|239, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|240, 1/2, -1/2\rangle = \left|3d_{xz}^{\beta} 3d_{yz}^{0} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|241, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|242, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{2} 3d_{z^2}^{0} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|243, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{2} 3d_{x^2-y^2}^{0}\right\rangle$$

$$|244, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{\beta} 3d_{z^2}^{0} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|245, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{2} 3d_{x^2-y^2}^{\beta}\right\rangle$$

$$|246, 1/2, -1/2\rangle = \left|3d_{xz}^{0} 3d_{yz}^{2} 3d_{xy}^{0} 3d_{z^2}^{\beta} 3d_{x^2-y^2}^{2}\right\rangle$$

$$|247,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0\right\rangle$$
$$|248,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2\right\rangle$$
$$|249,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|250,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$
$$|251,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$
$$|252,1/2,-1/2\rangle = \left|3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 25 45 65 85

$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 7, \frac{3}{2}, \frac{3}{2} \right\rangle = -\frac{1}{\sqrt{2}} I$
$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_x \right| 11, \frac{3}{2}, \frac{3}{2} \right\rangle = -\sqrt{\frac{3}{2}} I$

Δg_{xx}

$-\frac{1}{3}\zeta\Delta_{25-7}^{-1}$
$-\zeta\Delta_{25-11}^{-1}$

$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 26, \frac{3}{2}, \frac{3}{2} \right\rangle = -I$
$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 34, \frac{3}{2}, \frac{3}{2} \right\rangle = \sqrt{3} I$
$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_y \right| 35, \frac{3}{2}, \frac{3}{2} \right\rangle = I$

Δg_{yy}

$-\frac{2}{3}\zeta\Delta_{25-26}^{-1}$
$-2\zeta\Delta_{25-34}^{-1}$
$-\frac{2}{3}\zeta\Delta_{25-35}^{-1}$

$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 23, \frac{3}{2}, \frac{3}{2} \right\rangle = 2I$
$\left\langle 25, \frac{3}{2}, \frac{3}{2} \left| L_z \right| 29, \frac{3}{2}, \frac{3}{2} \right\rangle = I$

Δg_{zz}

$-\frac{8}{3}\zeta\Delta_{25-23}^{-1}$
$-\frac{2}{3}\zeta\Delta_{25-29}^{-1}$

For a multiplicity of 4 we have the following:

$D = \frac{1}{2}\left(\left\langle\frac{3}{2},\frac{3}{2}\middle|\frac{3}{2},\frac{3}{2}\right\rangle - \left\langle\frac{3}{2},\frac{1}{2}\middle|\frac{3}{2},\frac{1}{2}\right\rangle\right)$

$E = \frac{1}{\sqrt{3}}\left\langle\frac{3}{2},-\frac{1}{2}\middle|\frac{3}{2},\frac{3}{2}\right\rangle$

$$\begin{aligned}\left\langle\frac{3}{2},\frac{3}{2}\middle|\hat{H}_{eff}\middle|\frac{3}{2},\frac{3}{2}\right\rangle = -\frac{4}{9}\zeta^2 (\\ &+ \frac{1}{4}\Delta_1^{-1} + \frac{1}{40}\Delta_3^{-1} + \frac{1}{24}\Delta_8^{-1} + \frac{1}{72}\Delta_{12}^{-1} + \frac{1}{36}\Delta_{16}^{-1} + \frac{1}{60}\Delta_{20}^{-1} \\ &+ \Delta_{23}^{-1} + \frac{1}{4}\Delta_{29}^{-1} + \frac{1}{12}\Delta_{46}^{-1} + \frac{1}{4}\Delta_{54}^{-1} + \frac{1}{12}\Delta_{55}^{-1} + \frac{1}{16}\Delta_{103}^{-1} \\ &+ \frac{1}{48}\Delta_{105}^{-1} + \frac{1}{48}\Delta_{107}^{-1} + \frac{1}{144}\Delta_{109}^{-1} + \frac{1}{72}\Delta_{111}^{-1} + \frac{1}{8}\Delta_{116}^{-1} + \frac{3}{8}\Delta_{124}^{-1} \\ &+ \frac{1}{8}\Delta_{125}^{-1} + \frac{1}{24}\Delta_{156}^{-1} + \frac{1}{8}\Delta_{164}^{-1} + \frac{1}{24}\Delta_{165}^{-1} + \frac{1}{4}\Delta_{194}^{-1} + \frac{3}{4}\Delta_{195}^{-1} \\ &+ \frac{3}{4}\Delta_{203}^{-1} + \frac{1}{4}\Delta_{204}^{-1})\end{aligned}$$

$$\begin{aligned}\left\langle\frac{3}{2},\frac{1}{2}\middle|\hat{H}_{eff}\middle|\frac{3}{2},\frac{1}{2}\right\rangle = -\frac{4}{9}\zeta^2 (\\ &+ \frac{3}{20}\Delta_2^{-1} + \frac{3}{40}\Delta_4^{-1} + \frac{1}{24}\Delta_7^{-1} + \frac{1}{18}\Delta_9^{-1} + \frac{1}{72}\Delta_{11}^{-1} + \frac{5}{27}\Delta_{13}^{-1} \\ &+ \frac{1}{36}\Delta_{15}^{-1} + \frac{1}{27}\Delta_{17}^{-1} + \frac{1}{60}\Delta_{19}^{-1} + \frac{1}{45}\Delta_{21}^{-1} + \frac{1}{12}\Delta_{26}^{-1} + \frac{1}{4}\Delta_{34}^{-1} \\ &+ \frac{1}{12}\Delta_{35}^{-1} + \frac{1}{9}\Delta_{43}^{-1} + \frac{1}{36}\Delta_{49}^{-1} + \frac{1}{9}\Delta_{66}^{-1} + \frac{1}{3}\Delta_{74}^{-1} + \frac{1}{9}\Delta_{75}^{-1} \\ &+ \frac{1}{48}\Delta_{104}^{-1} + \frac{1}{144}\Delta_{106}^{-1} + \frac{1}{144}\Delta_{108}^{-1} + \frac{1}{432}\Delta_{110}^{-1} + \frac{1}{216}\Delta_{112}^{-1} + \frac{2}{3}\Delta_{113}^{-1} \\ &+ \frac{1}{6}\Delta_{119}^{-1} + \frac{1}{24}\Delta_{136}^{-1} + \frac{1}{8}\Delta_{144}^{-1} + \frac{1}{24}\Delta_{145}^{-1} + \frac{2}{9}\Delta_{153}^{-1} + \frac{1}{18}\Delta_{159}^{-1} \\ &+ \frac{1}{72}\Delta_{176}^{-1} + \frac{1}{24}\Delta_{184}^{-1} + \frac{1}{72}\Delta_{185}^{-1} + \frac{1}{12}\Delta_{224}^{-1} + \frac{1}{4}\Delta_{225}^{-1} + \frac{1}{4}\Delta_{233}^{-1} \\ &+ \frac{1}{12}\Delta_{234}^{-1})\end{aligned}$$

$$
\begin{aligned}
\left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = & -\frac{4}{9}\zeta^2 \, (\\
& -\frac{\sqrt{3}}{40}\Delta_{3}^{-1} + \frac{1}{12\sqrt{3}}\Delta_{8}^{-1} + \frac{1}{36\sqrt{3}}\Delta_{12}^{-1} + \frac{1}{18\sqrt{3}}\Delta_{16}^{-1} + \frac{1}{30\sqrt{3}}\Delta_{20}^{-1} \\
& -\frac{1}{3\sqrt{12}}\Delta_{46}^{-1} - \frac{1}{\sqrt{12}}\Delta_{54}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{55}^{-1} - \frac{\sqrt{3}}{48}\Delta_{103}^{-1} - \frac{\sqrt{3}}{144}\Delta_{105}^{-1} \\
& -\frac{\sqrt{3}}{144}\Delta_{107}^{-1} - \frac{1}{144\sqrt{3}}\Delta_{109}^{-1} - \frac{1}{72\sqrt{3}}\Delta_{111}^{-1} + \frac{\sqrt{3}}{24}\Delta_{116}^{-1} + \frac{\sqrt{3}}{8}\Delta_{124}^{-1} \\
& +\frac{\sqrt{3}}{24}\Delta_{125}^{-1} + \frac{\sqrt{3}}{72}\Delta_{156}^{-1} + \frac{\sqrt{3}}{24}\Delta_{164}^{-1} + \frac{\sqrt{3}}{72}\Delta_{165}^{-1} - \frac{\sqrt{3}}{12}\Delta_{194}^{-1} \\
& -\frac{\sqrt{3}}{4}\Delta_{195}^{-1} - \frac{\sqrt{3}}{4}\Delta_{203}^{-1} - \frac{\sqrt{3}}{12}\Delta_{204}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
D = & \frac{2}{9}\zeta^2 \, (\\
& -\frac{1}{4}\Delta_{1}^{-1} + \frac{3}{20}\Delta_{2}^{-1} - \frac{1}{40}\Delta_{3}^{-1} + \frac{3}{40}\Delta_{4}^{-1} + \frac{1}{24}\Delta_{7}^{-1} - \frac{1}{24}\Delta_{8}^{-1} \\
& +\frac{1}{18}\Delta_{9}^{-1} + \frac{1}{72}\Delta_{11}^{-1} - \frac{1}{72}\Delta_{12}^{-1} + \frac{5}{27}\Delta_{13}^{-1} + \frac{1}{36}\Delta_{15}^{-1} - \frac{1}{36}\Delta_{16}^{-1} \\
& +\frac{1}{27}\Delta_{17}^{-1} + \frac{1}{60}\Delta_{19}^{-1} - \frac{1}{60}\Delta_{20}^{-1} + \frac{1}{45}\Delta_{21}^{-1} - \Delta_{23}^{-1} + \frac{1}{12}\Delta_{26}^{-1} \\
& -\frac{1}{4}\Delta_{29}^{-1} + \frac{1}{4}\Delta_{34}^{-1} + \frac{1}{12}\Delta_{35}^{-1} + \frac{1}{9}\Delta_{43}^{-1} - \frac{1}{12}\Delta_{46}^{-1} + \frac{1}{36}\Delta_{49}^{-1} \\
& -\frac{1}{4}\Delta_{54}^{-1} - \frac{1}{12}\Delta_{55}^{-1} + \frac{1}{9}\Delta_{66}^{-1} + \frac{1}{3}\Delta_{74}^{-1} + \frac{1}{9}\Delta_{75}^{-1} - \frac{1}{16}\Delta_{103}^{-1} \\
& +\frac{1}{48}\Delta_{104}^{-1} - \frac{1}{48}\Delta_{105}^{-1} + \frac{1}{144}\Delta_{106}^{-1} - \frac{1}{48}\Delta_{107}^{-1} + \frac{1}{144}\Delta_{108}^{-1} - \frac{1}{144}\Delta_{109}^{-1} \\
& +\frac{1}{432}\Delta_{110}^{-1} - \frac{1}{72}\Delta_{111}^{-1} + \frac{1}{216}\Delta_{112}^{-1} + \frac{2}{3}\Delta_{113}^{-1} - \frac{1}{8}\Delta_{116}^{-1} + \frac{1}{6}\Delta_{119}^{-1} \\
& -\frac{3}{8}\Delta_{124}^{-1} - \frac{1}{8}\Delta_{125}^{-1} + \frac{1}{24}\Delta_{136}^{-1} + \frac{1}{8}\Delta_{144}^{-1} + \frac{1}{24}\Delta_{145}^{-1} + \frac{2}{9}\Delta_{153}^{-1} \\
& -\frac{1}{24}\Delta_{156}^{-1} + \frac{1}{18}\Delta_{159}^{-1} - \frac{1}{8}\Delta_{164}^{-1} - \frac{1}{24}\Delta_{165}^{-1} + \frac{1}{72}\Delta_{176}^{-1} + \frac{1}{24}\Delta_{184}^{-1} \\
& +\frac{1}{72}\Delta_{185}^{-1} - \frac{1}{4}\Delta_{194}^{-1} - \frac{3}{4}\Delta_{195}^{-1} - \frac{3}{4}\Delta_{203}^{-1} - \frac{1}{4}\Delta_{204}^{-1} + \frac{1}{12}\Delta_{224}^{-1} \\
& +\frac{1}{4}\Delta_{225}^{-1} + \frac{1}{4}\Delta_{233}^{-1} + \frac{1}{12}\Delta_{234}^{-1} \,)
\end{aligned}
$$

$$
\begin{aligned}
E =& \frac{4}{9\sqrt{3}}\zeta^2 \, (\\
& + \frac{\sqrt{3}}{40}\Delta_3^{-1} - \frac{1}{12\sqrt{3}}\Delta_8^{-1} - \frac{1}{36\sqrt{3}}\Delta_{12}^{-1} - \frac{1}{18\sqrt{3}}\Delta_{16}^{-1} - \frac{1}{30\sqrt{3}}\Delta_{20}^{-1} \\
& + \frac{1}{3\sqrt{12}}\Delta_{46}^{-1} + \frac{1}{\sqrt{12}}\Delta_{54}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{55}^{-1} + \frac{\sqrt{3}}{48}\Delta_{103}^{-1} + \frac{\sqrt{3}}{144}\Delta_{105}^{-1} \\
& + \frac{\sqrt{3}}{144}\Delta_{107}^{-1} + \frac{1}{144\sqrt{3}}\Delta_{109}^{-1} + \frac{1}{72\sqrt{3}}\Delta_{111}^{-1} - \frac{\sqrt{3}}{24}\Delta_{116}^{-1} - \frac{\sqrt{3}}{8}\Delta_{124}^{-1} \\
& - \frac{\sqrt{3}}{24}\Delta_{125}^{-1} - \frac{\sqrt{3}}{72}\Delta_{156}^{-1} - \frac{\sqrt{3}}{24}\Delta_{164}^{-1} - \frac{\sqrt{3}}{72}\Delta_{165}^{-1} + \frac{\sqrt{3}}{12}\Delta_{194}^{-1} \\
& + \frac{\sqrt{3}}{4}\Delta_{195}^{-1} + \frac{\sqrt{3}}{4}\Delta_{203}^{-1} + \frac{\sqrt{3}}{12}\Delta_{204}^{-1} \,)
\end{aligned}
$$